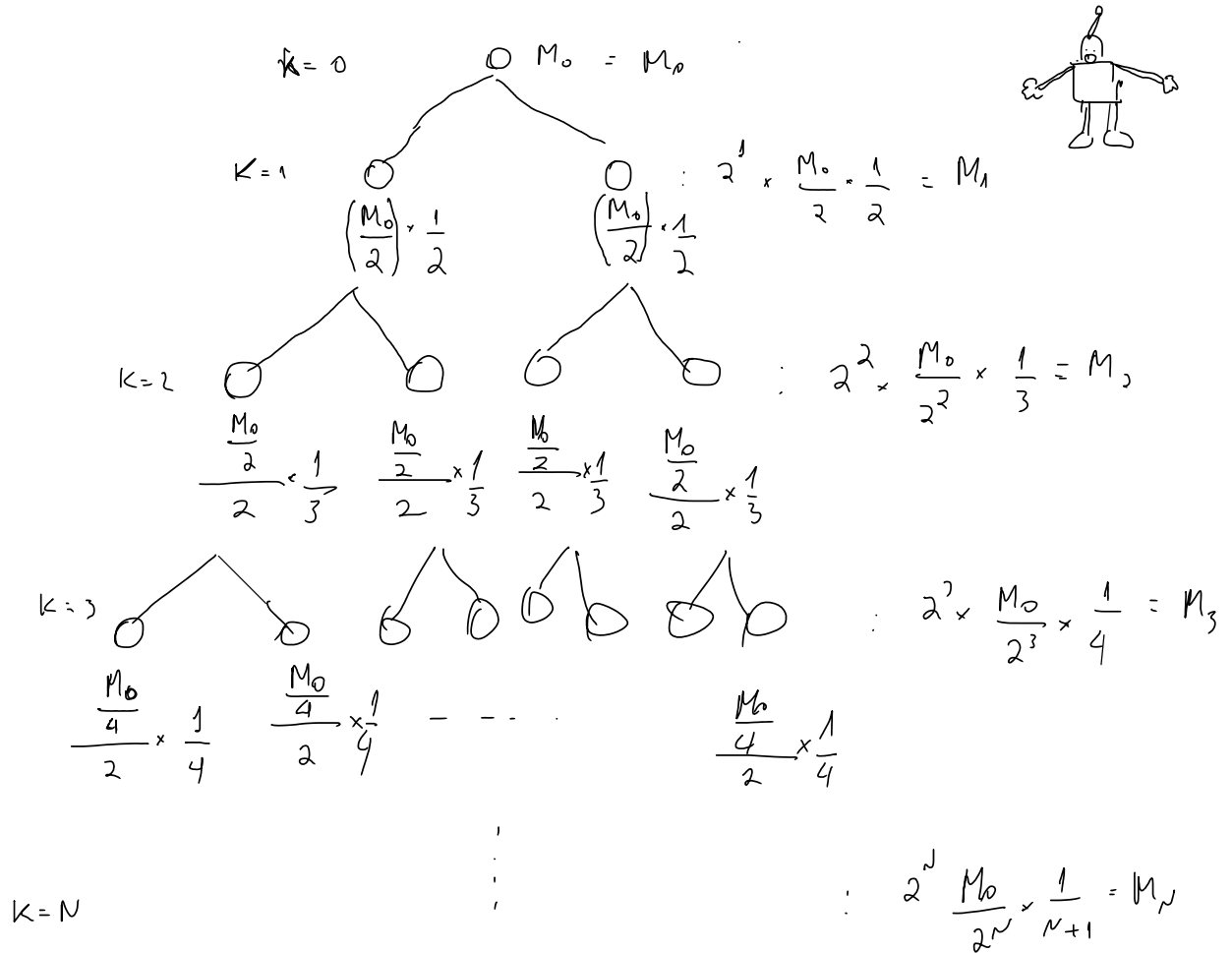


$$M = \sum_{k=1}^N 2^k \left[\frac{M_0}{2^k(k+1)} \right] \quad \therefore \lim_{N \rightarrow \infty} M(N) \rightarrow \infty \quad (\text{diverge})$$



$$M(N) = M_0 + M_1 + M_2 + M_3 + \dots + M_N$$

$$M(N) = M_0 + \frac{M_0}{2^1} \times \frac{1}{2} + 2^2 \times \frac{M_0}{2^2} \times \frac{1}{3} + 2^3 \times \frac{M_0}{2^3} \times \frac{1}{4} + \dots + 2^N \times \frac{M_0}{2^N} \times \frac{1}{N+1} = \sum_{k=1}^N 2^k \left[\frac{M_0}{2^k(k+1)} \right]$$

$$M(N) = \sum_{k=0}^N \frac{M_0}{k+1}$$

$$\sum_{k=0}^N \frac{1}{k+1} = \sum_{k'=1}^N \frac{1}{k'} \rightarrow \text{SÉRIE HARMÔNICAS} \quad (\text{divergente})$$

$$k+1 \rightarrow k'$$

$$k=0 \Rightarrow k'-1=0 \Rightarrow k'=1$$

$$\zeta(s) = \sum_{k=1}^{\infty} \frac{1}{k^s} \rightarrow \text{função } \zeta \text{ DE RIEMANN}$$